

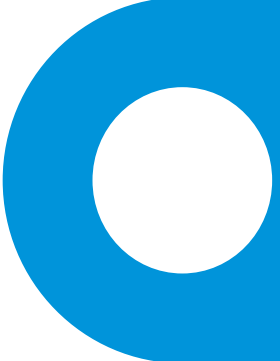
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Faculty of Computer Science and Engineering



Software Testing Strategies



Software Testing (1)

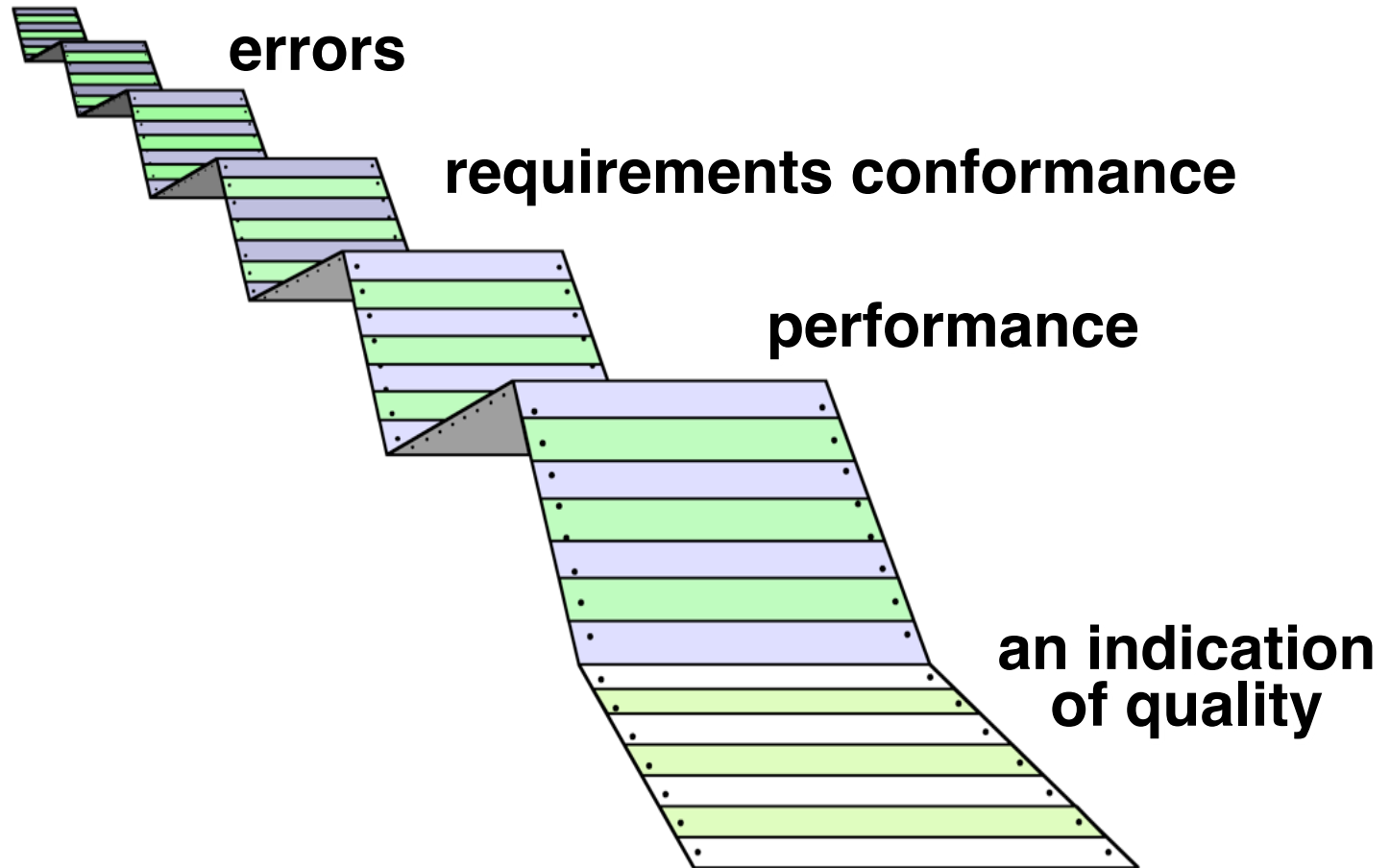
Testing is the process of exercising a program with the specific intent of finding errors prior to delivery to the end user. Testing is intended to show that a program does what it is intended to do.

Software Testing (2)

- Due to human inability to work and communicate perfectly (without errors), the software must be checked
- Software testing is a quality control function that has one primary goal—to find errors. The job of software quality assurance is to ensure that testing is properly planned and efficiently conducted so that it has the highest likelihood of achieving its primary goal.
- Testing often accounts for more project effort than any other software engineering action



What Testing Shows



Strategic Approach

- To perform effective testing, you should conduct effective technical reviews. By doing this, many errors will be eliminated before testing commences.
- Testing begins at the component level and works "outward" toward the integration of the entire computer-based system.
- Different testing techniques are appropriate for different software engineering approaches and at different points in time.
- Testing is conducted by the developer of the software and (for large projects) an independent test group.
- Testing and debugging are different activities, but debugging must be accommodated in any testing strategy.

V & V

- *Verification* refers to the set of tasks that ensure that software correctly implements a specific function.
- *Validation* refers to a different set of tasks that ensure that the software that has been built is traceable to customer requirements. Boehm [Boe81] states this another way:
 - *Verification: "Are we building the product right?"*
 - *Validation: "Are we building the right product?"*



Who Tests the Software?



developer

**Understands the system
but, will test "gently"
and, is driven by "delivery"**

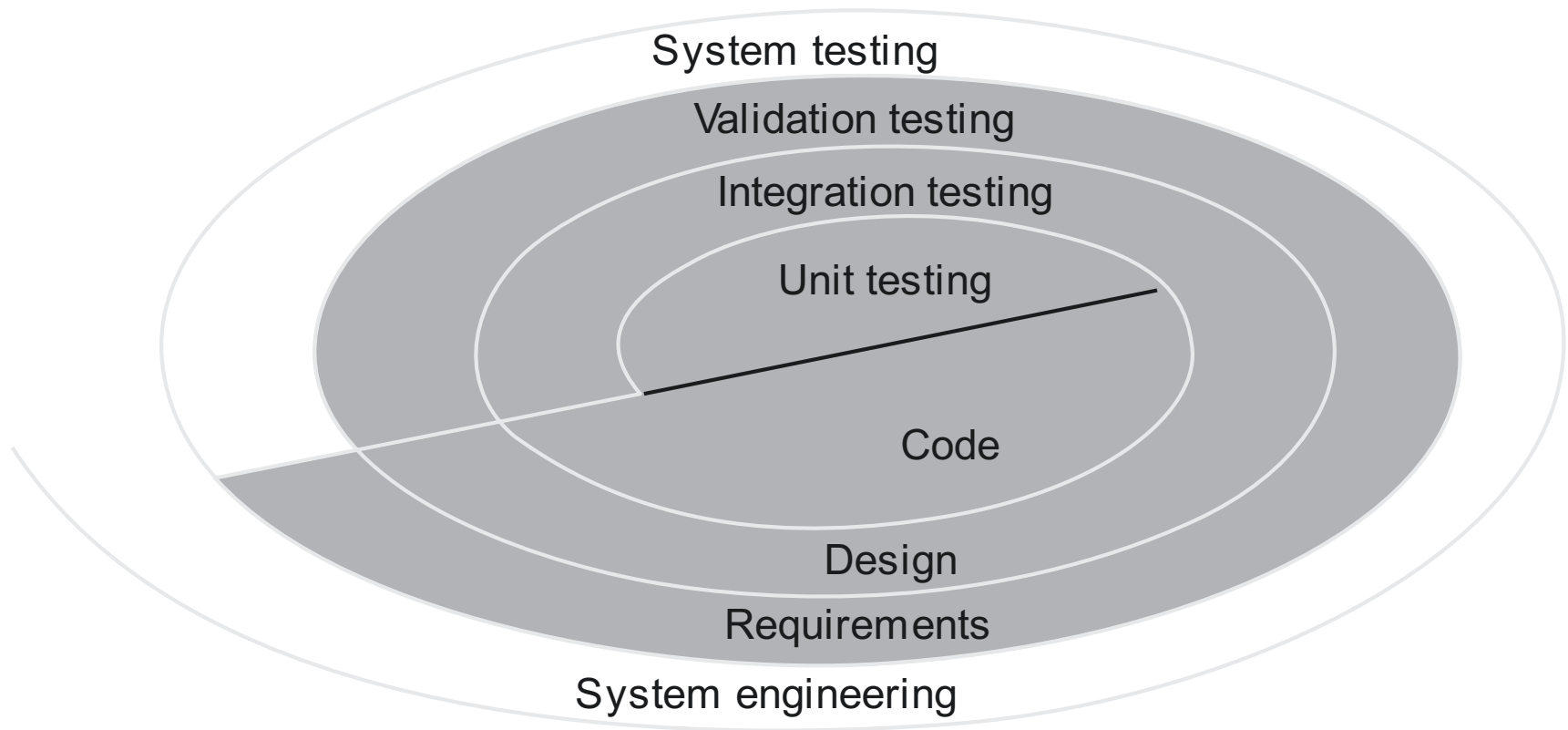


independent tester

**Must learn about the system,
but, will attempt to break it
and, is driven by quality**

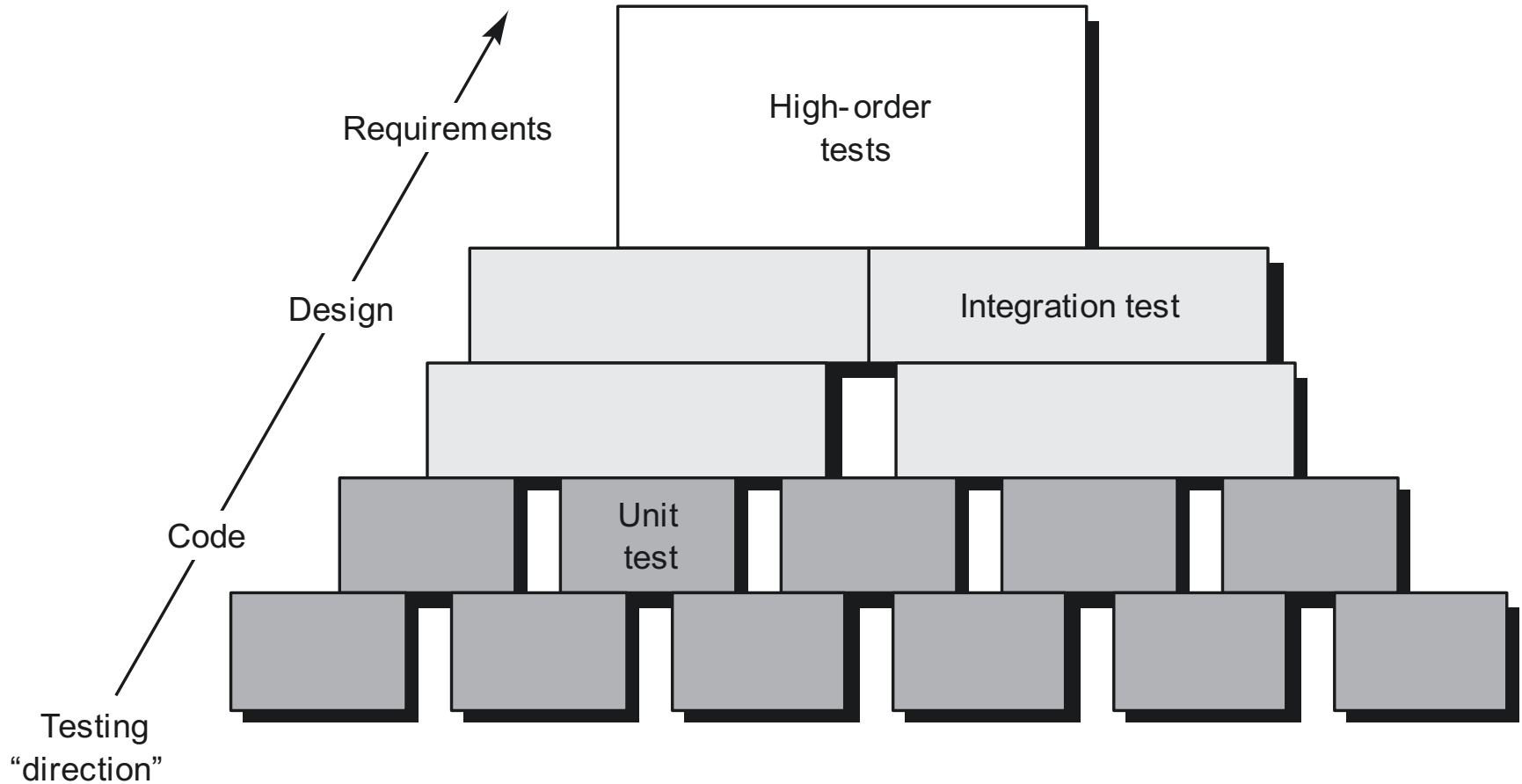


Testing Strategy





Software testing steps

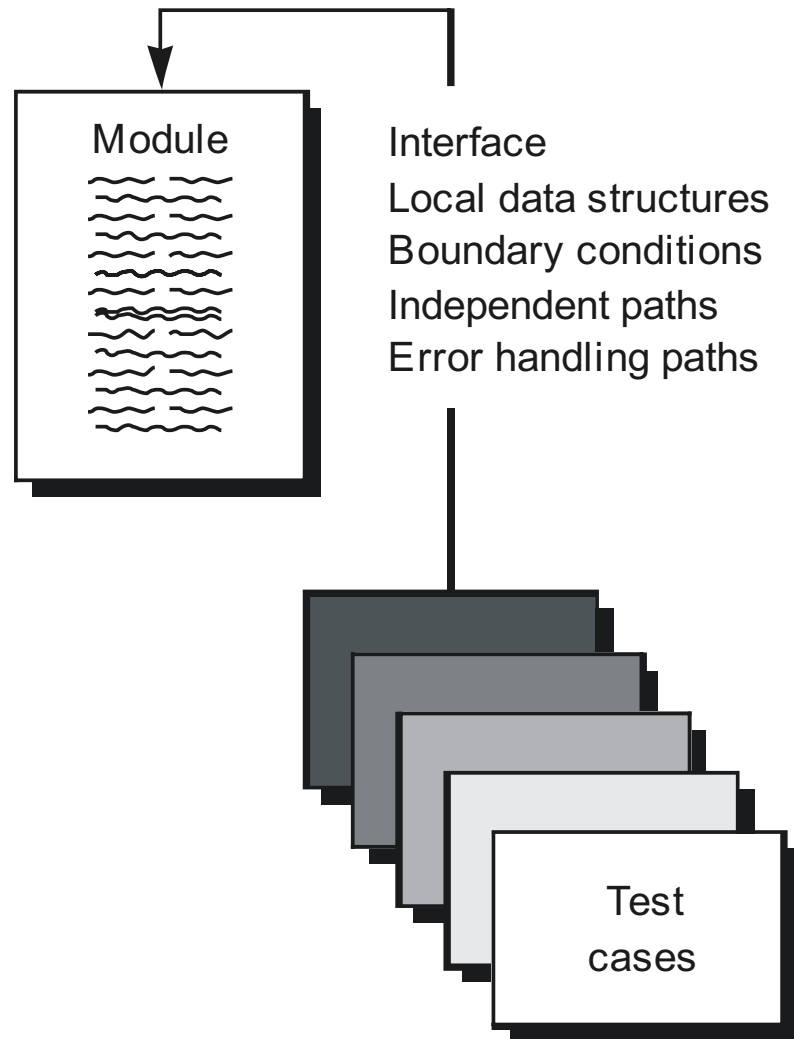




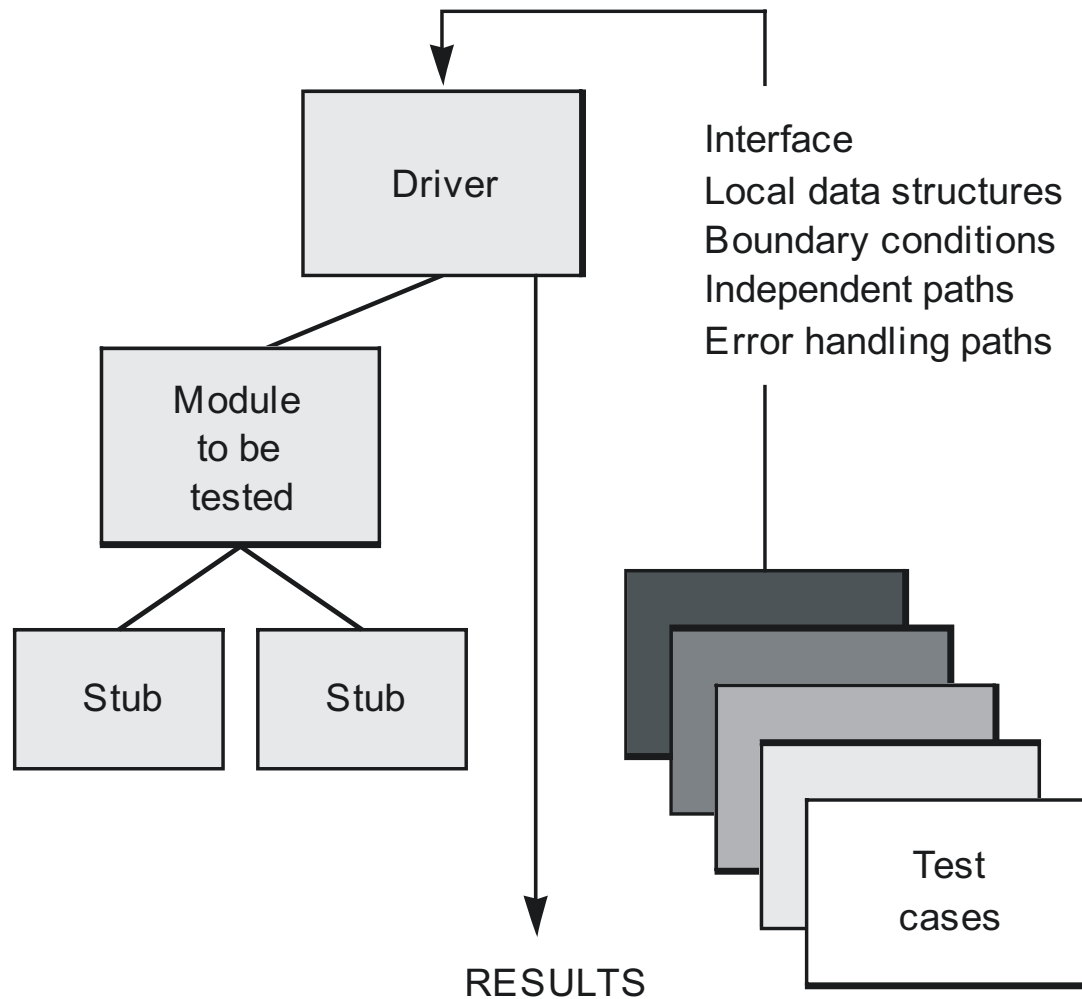
Strategic Issues

- Specify product requirements in a quantifiable manner long before testing commences.
- State testing objectives explicitly.
- Understand the users of the software and develop a profile for each user category.
- Develop a testing plan that emphasizes “rapid cycle testing.”
- Build “robust” software that is designed to test itself
- Use effective technical reviews as a filter prior to testing
- Conduct technical reviews to assess the test strategy and test cases themselves.
- Develop a continuous improvement approach for the testing process.

Unit Testing



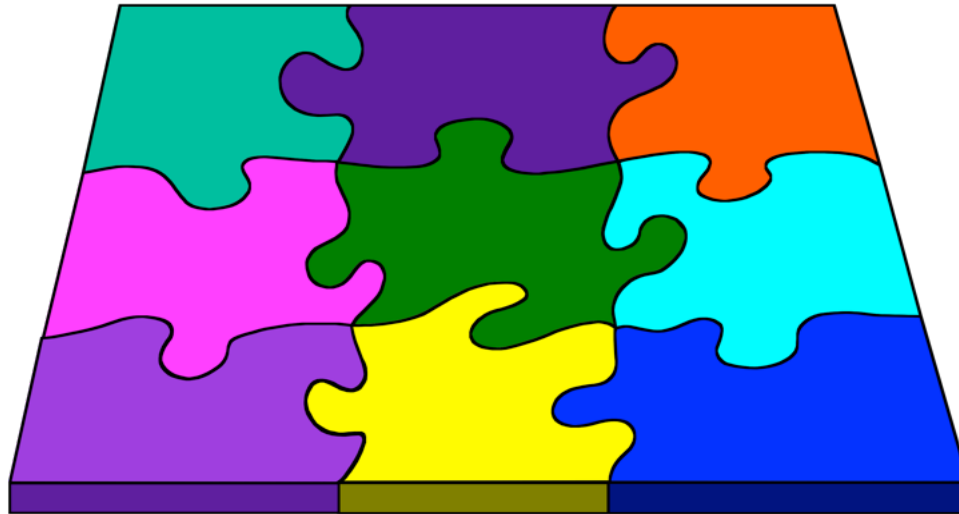
Unit Test Environment



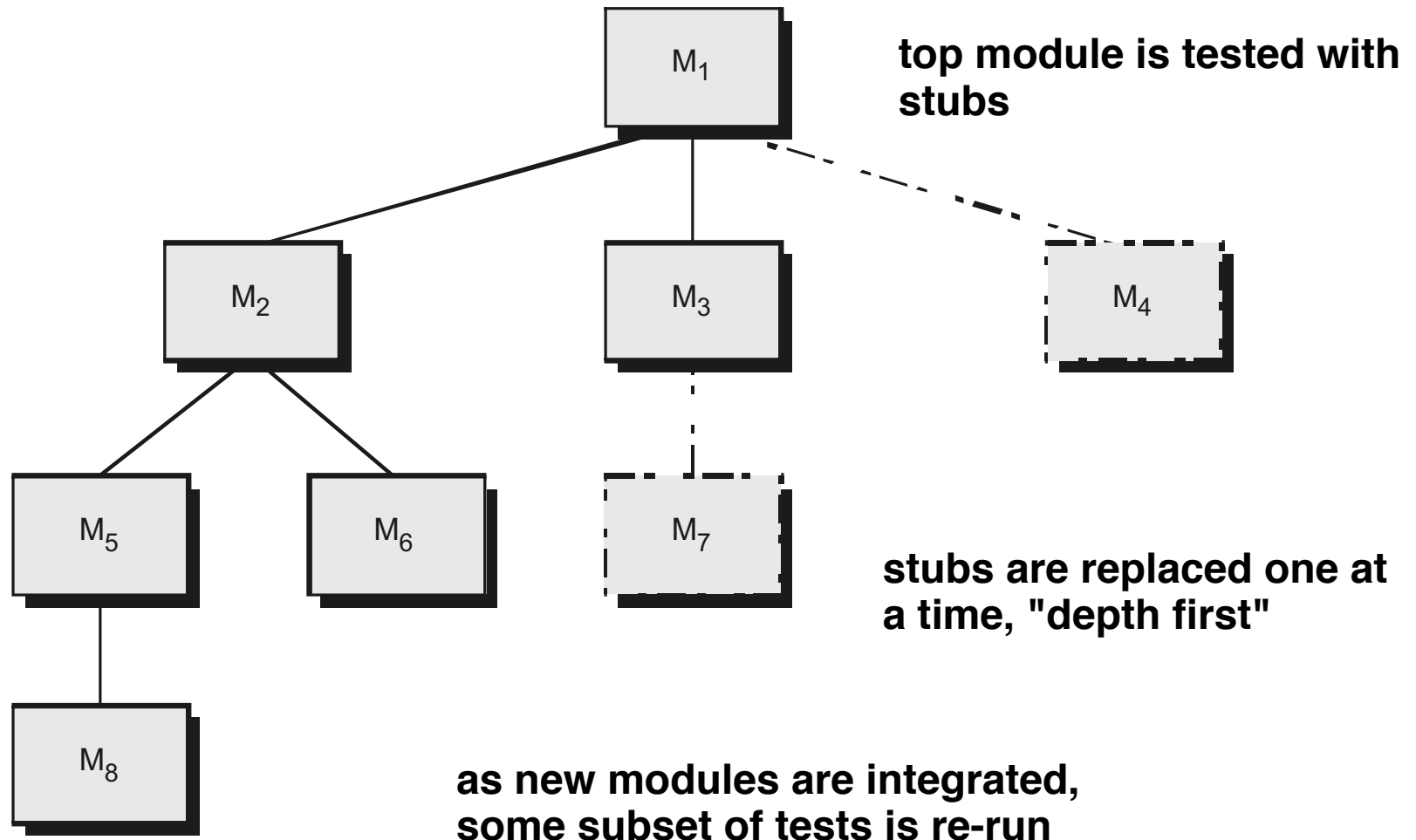
Integration Testing Strategies

Options:

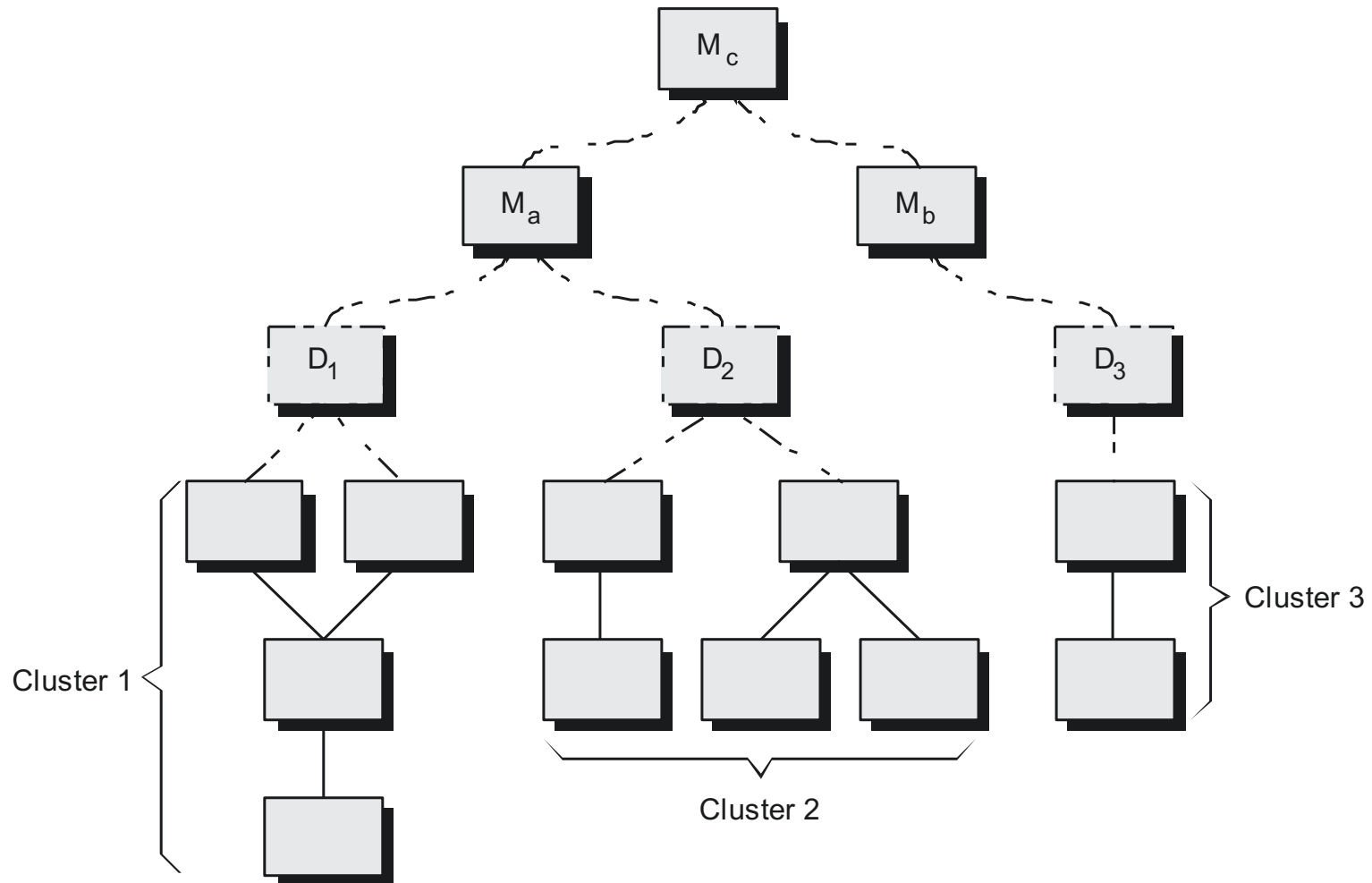
- the “big bang” approach
- an incremental construction strategy



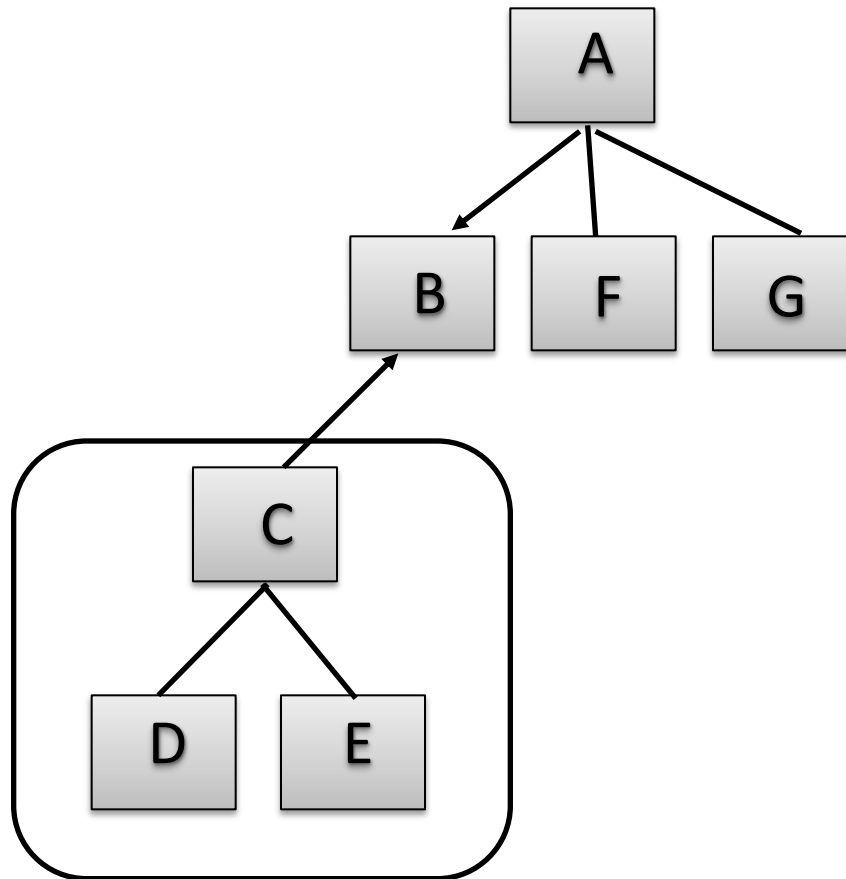
Top Down Integration



Bottom-up Integration



Sandwich Testing



Regression Testing

- *Regression testing* is the re-execution of some subset of tests that have already been conducted to ensure that changes have not propagated unintended side effects
- Whenever software is corrected, some aspect of the software configuration (the program, its documentation, or the data that support it) is changed.
- Regression testing helps to ensure that changes (due to testing or for other reasons) do not introduce unintended behavior or additional errors.
- Regression testing may be conducted manually, by re-executing a subset of all test cases or using automated capture/playback tools.



Smoke Testing

- A common approach for creating “daily builds” for product software
- Smoke testing steps:
 - Software components that have been translated into code are integrated into a “build.”
 - A build includes all data files, libraries, reusable modules, and engineered components that are required to implement one or more product functions.
 - A series of tests is designed to expose errors that will keep the build from properly performing its function.
 - The intent should be to uncover “show stopper” errors that have the highest likelihood of throwing the software project behind schedule.
 - The build is integrated with other builds and the entire product (in its current form) is smoke tested daily.
 - The integration approach may be top down or bottom up.

Smoke Testing Benefits

- Smoke testing provides a number of benefits when it is applied on complex, timecritical software engineering projects:
- *Integration risk is minimized.*
- *The quality of the end-product is improved.*
- *Error diagnosis and correction are simplified.*
- *Progress is easier to assess.*

Integration Test Documentation

- An overall plan for integration of the software and a description of specific tests is documented in a *Test Specification*
- This work product incorporates a test plan and a test procedure and becomes part of the software configuration.
- Test phases criteria:
 - ☐ Interface integrity
 - ☐ Functional validity
 - ☐ Information content
 - ☐ Performance

Object-Oriented Testing

- Begins by evaluating the correctness and consistency of the analysis and design models
- Testing strategy changes
- Test case design draws on conventional methods, but also encompasses special features

00 Testing Strategy

- class testing is the equivalent of unit testing
 - operations within the class are tested
 - the state behavior of the class is examined
- integration applied three different strategies
 - thread-based testing—integrates the set of classes required to respond to one input or event
 - use-based testing—integrates the set of classes required to respond to one use case
 - cluster testing—integrates the set of classes required to demonstrate one collaboration

Validation Testing

- Begins at the culmination of integration testing
- Validation succeeds when software functions in a manner that can be reasonably expected by the customer
 - focuses on user-visible actions and user-recognizable output from the system (*Software Requirements Specification*)
- Alpha and Beta Testing



System testing

- System testing

- Focus is on system integration

- Recovery testing

- forces the software to fail in a variety of ways and verifies that recovery is properly performed

- Security testing

- verifies that protection mechanisms built into a system will, in fact, protect it from improper penetration

- Stress testing

- executes a system in a manner that demands resources in abnormal quantity, frequency, or volume

- Performance Testing

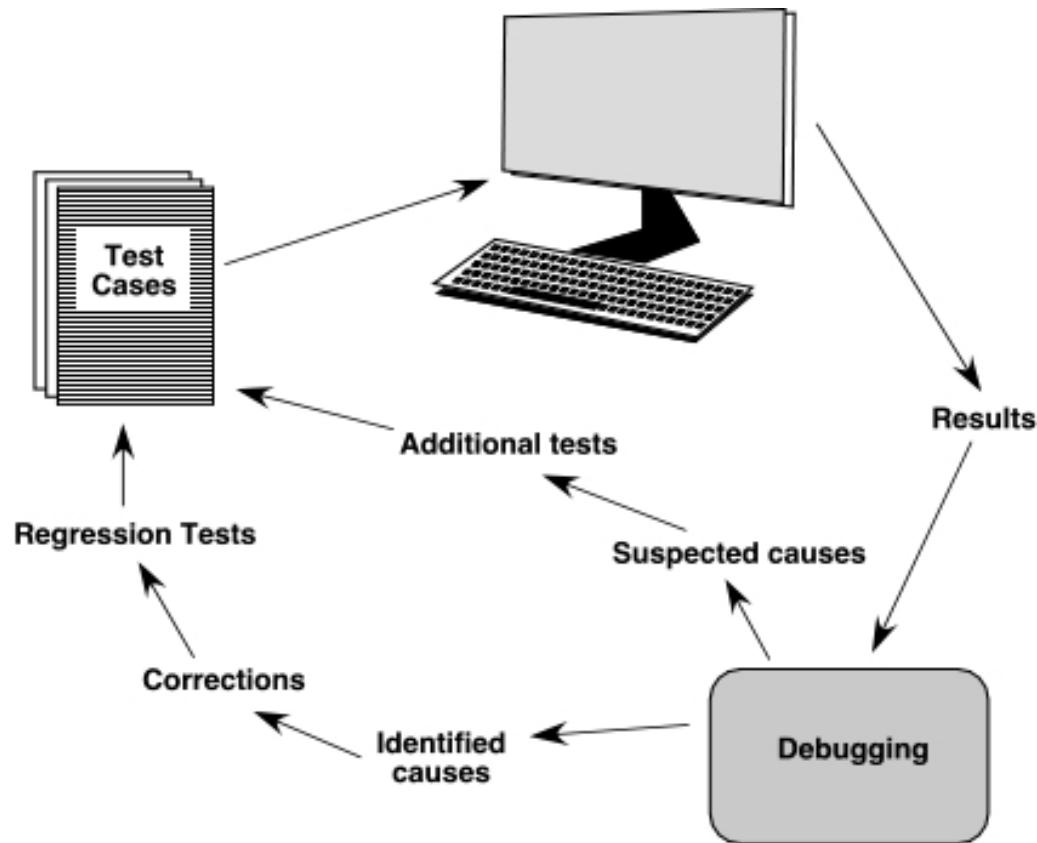
- test the run-time performance of software within the context of an integrated system

- Deployment Testing

Debugging: A Diagnostic Process

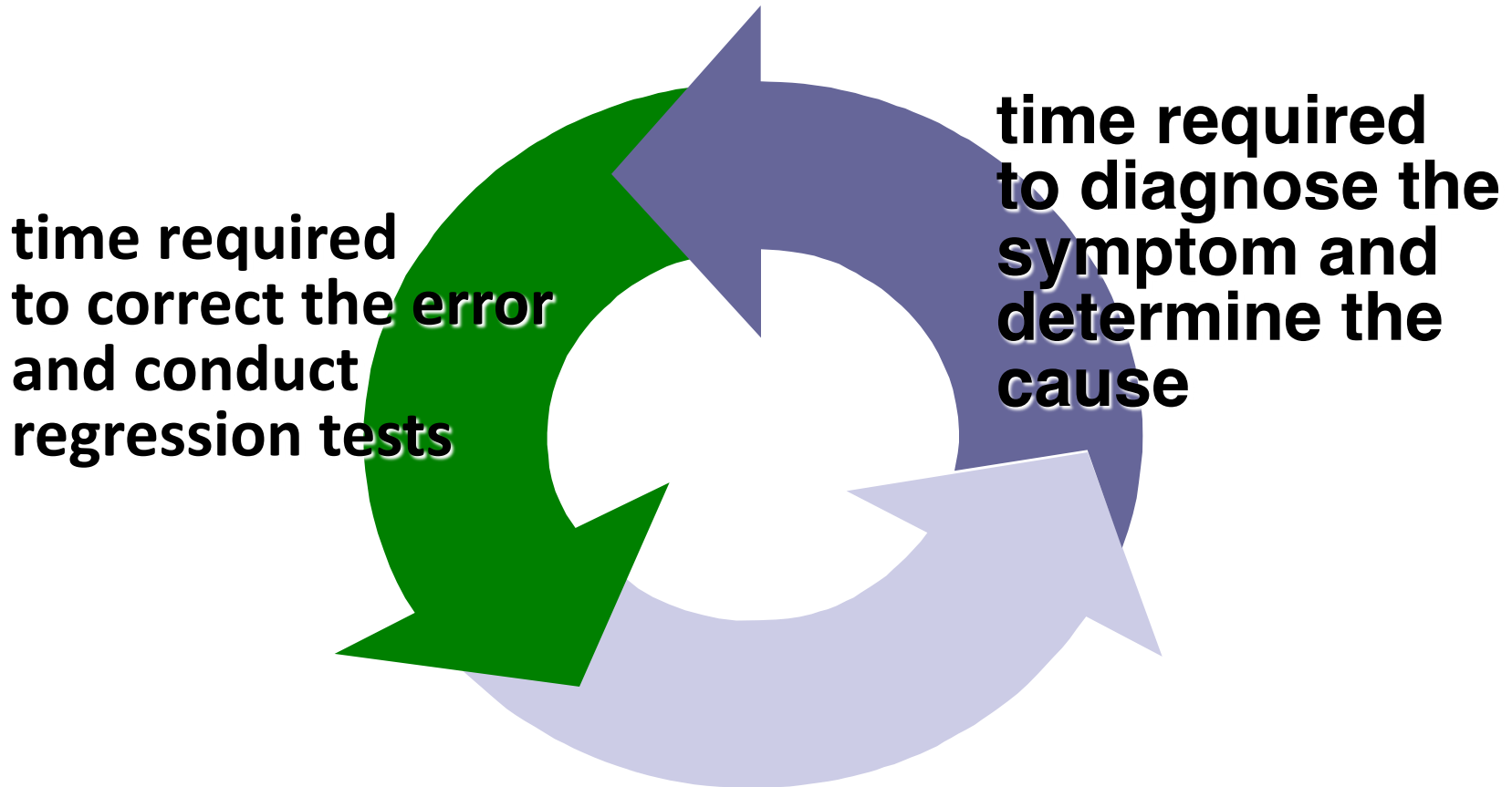


The Debugging Process

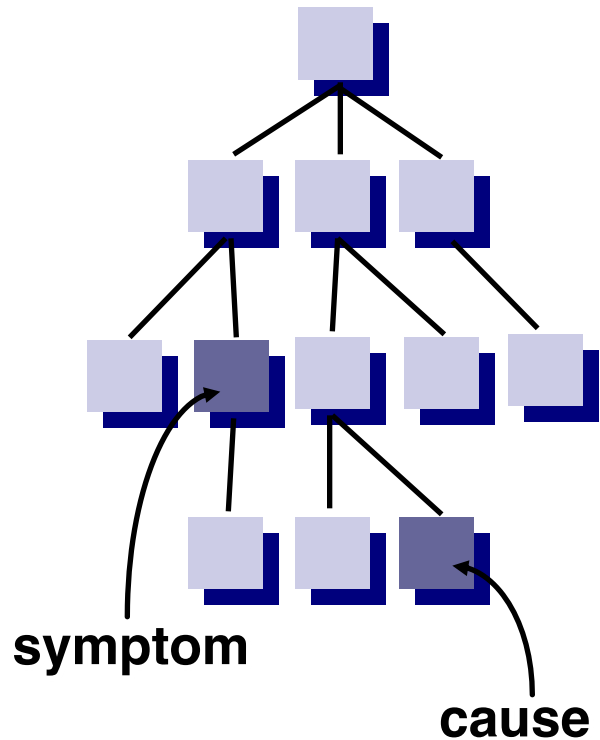




Debugging Effort

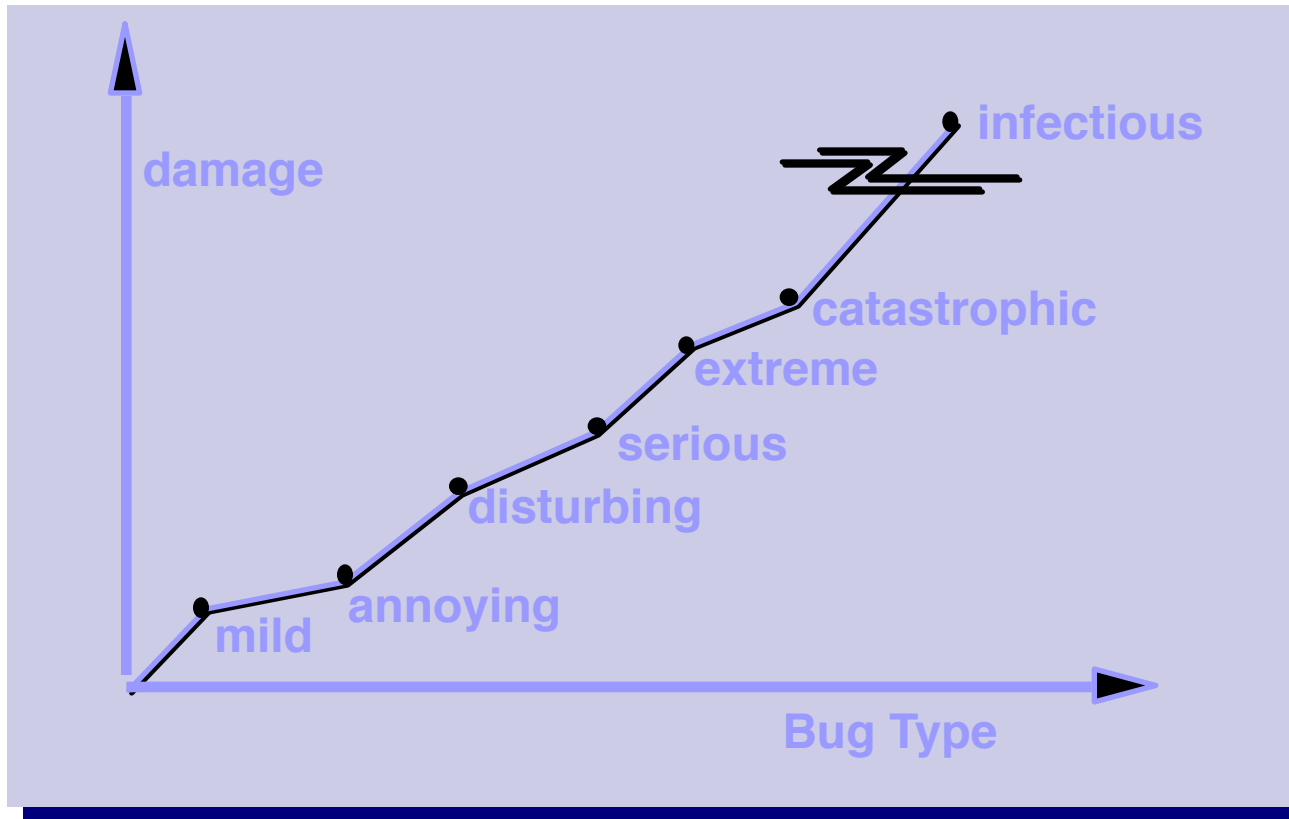


Symptoms & Causes



- ❑ symptom and cause may be geographically separated
- ❑ symptom may disappear when another problem is fixed
- ❑ cause may be due to a combination of non-errors
- ❑ It may be difficult to accurately reproduce input conditions
- ❑ symptom may be caused by human error that is not easily traced.
- ❑ symptom may be a result of timing problems, rather than processing problems.

Consequences of Bugs



Bug Categoriesfunction-related bugs,
system-related bugs, data bugs, coding bugs,
design bugs, documentation bugs, standards
violations, etc.



Debugging Techniques

- ❏ brute force / testing
- ❏ backtracking
- ❏ induction
- ❏ deduction

Correcting the Error

- *Is the cause of the bug reproduced in another part of the program?* In many situations, a program defect is caused by an erroneous pattern of logic that may be reproduced elsewhere.
- *What "next bug" might be introduced by the fix I'm about to make?* Before the correction is made, the source code (or, better, the design) should be evaluated to assess coupling of logic and data structures.
- *What could we have done to prevent this bug in the first place?* This question is the first step toward establishing a statistical software quality assurance approach. If you correct the process as well as the product, the bug will be removed from the current program and may be eliminated from all future programs.

Key points

- When testing software, you should try to ‘break’ the software by using experience and guidelines to choose types of test case that have been effective in discovering defects in other systems.
- *Verification: "Are we building the product right?"*
- *Validation: "Are we building the right product?"*
- *Unit testing* focuses verification effort on the smallest unit of software design
- Integration testing is a systematic technique for constructing the software architecture while at the same time conducting tests to uncover errors associated with interfacing.

Key points

- *System testing* is a series of different tests whose primary purpose is to fully exercise the computer-based system.
- When a test case uncovers an error, debugging is the process that results in the removal of the error.
- *Think* -- before you act to correct
- Once you correct the bug, use regression testing to uncover any side effects